

Day 20 – Managed Identities in Microsoft Entra ID

Objective

Learn how Azure services use **Managed Identities** to securely access Azure resources without storing usernames, passwords, or secrets.

Skills You'll Learn

- System-assigned Managed Identity
- User-assigned Managed Identity
- Azure Key Vault authentication
- Secretless authentication
- Service-to-service authentication

What is a Managed Identity?

A **Managed Identity** is an automatically managed identity in Microsoft Entra ID that Azure services use to authenticate to other Azure services securely.

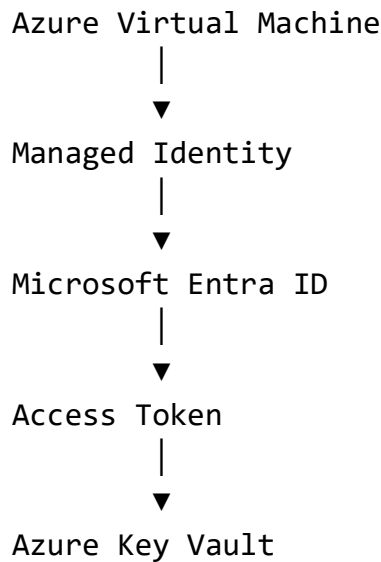
Instead of storing passwords or API keys, Azure automatically handles authentication.

Types of Managed Identities

Type	Description
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System-assigned	Created automatically for a single Azure resource and deleted when that resource is deleted.
User-assigned	Created as a standalone Azure resource and can be attached to multiple Azure resources.

Authentication Flow



Real-World Example

A virtual machine needs to retrieve a database password stored in **Azure Key Vault**.

Without Managed Identity:

- Store a password or secret in the VM (less secure).

With Managed Identity:

- The VM requests an access token from Microsoft Entra ID.
- Microsoft Entra ID verifies the VM's identity.
- Azure Key Vault returns the secret.
- No password or secret is stored on the VM.

